Time: 1 hr.
Paper Code: CSE–501

B. Tech. (5th Semester) 2nd Sessional Test 2025-26
Introduction to Data Mining

Max. Marks: 12

Note: Attempt All Question

Q1. CO3 Consider the training examples shown below for classification problem. [4] [BL-4,5]

(i) What is the entropy of this collection of training examples with respect to the positive class?

(ii) Calculate the information gains and which is the best split according to the classification error rate?

Instance	A1	A2	A3	Class
1.	T	T	3	+
2.	T	T	6	+
3.	T	F	5	-
4.	F	F	4	+
5.	F	T	7	-
6.	F	T	4	+
7.	F	F	8	-
8.	T	F	7	+
9.	F	T	6	+
10.	F	F	9	-

Q2. CO3 The following dataset is given, and you have to build a model using Naïve Bayes algorithm to determine the species based on the attributes. [4] [BL-4]

S. No	Color	Leg	Height	Smell	Species
1.	White	3	Short	Yes	M
2.	Green	2	Tall	No	M
3.	Green	3	Short	Yes	M
4.	White	3	Short	Yes	M
5.	Green	2	Short	No	H
6.	White	2	Tall	No	H
7.	White	2	Tall	No	H
8.	White	2	Short	Yes	H

Using the above data, we have to identify the species of an entity with the following attributes.
X= (Color= "Green", Legs=2, Height=Tall, Smell =No)
Determine the solution using Naïve Bayes classifier.

Q3. CO4 (i) The support vector machine is highly accurate classification method. However, SVM classifiers suffer from slow processing when training with a large set of data tuples. Discuss how to overcome this difficulty and develop a scalable SVM algorithm for efficient SVM classification in large data sets. [3+1=4] [BL-3,2]

(ii) Compare the advantages and disadvantages of eager classification versus lazy classification.